A universal physics-informed neural network for used-LFP RUL:

50-cycle characterisation across three manufacturers, 8× cycling reduction vs pure PyBaMM

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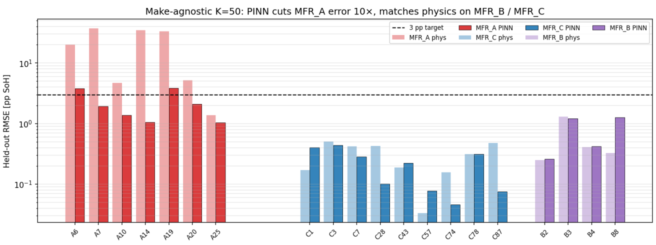
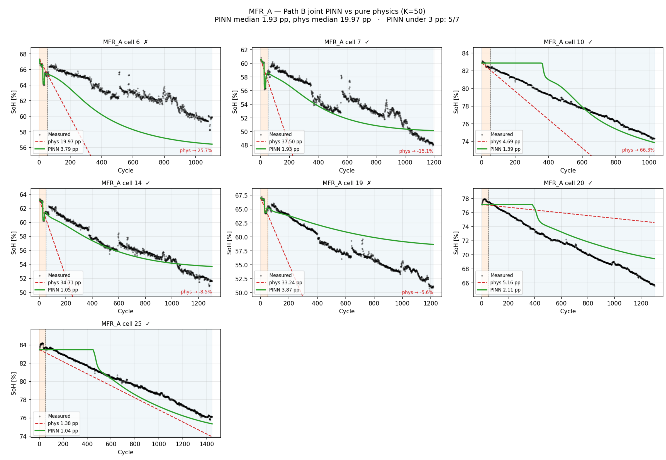
Motivation. Second-life repurposing of used LFP cells for stationary energy storage is bottlenecked by characterisation cost: pack builders must estimate remaining useful life (RUL) with no beginning-of-life data and limited cycling. Pure PyBaMM [1] reaction-limited SEI calibration requires ≈400 cycles per cell for < 3 pp SoH extrapolation error on used cells. Prior second-life optimisation [3] and universal-differential-equation degradation [4] work improves calibration quality without reducing the cycling budget. We ask: *can one physics-informed neural network with fixed hyperparameters cut required characterisation across LFP manufacturers?*

Method. A joint PINN represents SoH as $SoH_\theta(n) = SoH_{init} - softplus(NN_\theta)$ (hard-monotonic decrement) with a per-cell latent embedding. Physics constraint: $dSoH/dn = -k_{SEI}(i)\,SoH^{p(i)}$ (Ramadass-canonical rxn-limited SEI [5]), with per-cell learnable (k_{SEI}, p) warm-started from a linear fit on the training window. Loss combines data MSE, physics MSE on collocation points across the full cycle domain, boundary anchoring, and monotonicity regularisation. Adam + cosine schedule, $\lambda_{phys}=2$, 10k epochs, ≈4 min on a single GPU.

Result. 20 LFP cells across three manufacturers (7 MFR_A, 9 MFR_C, 4 MFR_B), *identical* architecture and hyperparameters (K=50, ≈68k params): the joint PINN achieves held-out RMSE < 3 pp on **18/20 cells** (Fig. 1a). On deep-second-life MFR_A cells (15–25 pp fade over 1000+ cy) where pure physics fails catastrophically (median 20 pp), the PINN maintains **median 1.93 pp** — a **10× reduction** and 7/7 head-to-head wins (Fig. 1b). On near-fresh MFR_B/MFR_C cells (1–3 pp fade) physics is already near-optimal (< 0.4 pp); the PINN matches without harm. **Vs the pure-physics K=400 baseline, the same PINN reaches equivalent engineering accuracy at K=50 — 8× cycling reduction with make-agnostic transfer.**



(a) Per-cell held-out RMSE across 3 manufacturers (log scale). MFR_A physics towers above 3 pp target; PINN matched or below across all makes.



(b) MFR_A trajectories (K=50). Physics dashed lines exit the frame with impossible-SoH annotations; PINN green tracks measured.

Scope & PyBaMM usage. Results hold for LFP at 25 °C isothermal. Cells excluded by upstream shape filter (non-monotonic, three-regime, batch-transition data artefacts) remain out of scope — PINN cannot recover cells whose canonical trajectories don't correspond to a coherent lifetime. PyBaMM `v26.4.3` provides the reference chemistry (`Prada2013`); its analytical rxn-limited SEI rate is embedded as the physics constraint.

References. [1] V. Sulzer et al., *PyBaMM*, J. Open Res. Softw. 9(1) 14–21 (2021). [2] M. Prada et al., *$LiFePO_4$-graphite Li-ion model*, J. Electrochem. Soc. 160(4) A616–A628 (2013). [3] O. Ouledboutaarija et al., *Advancing Second-Life Batteries*, PyBaMM Conf. 2025, p. 19. [4] J. A. Kuzhiyil et al., *Generalisability via UDE*, PyBaMM Conf. 2025, p. 12. [5] P. Ramadass et al., *Capacity fade first-principles model*, J. Electrochem. Soc. 151(2) A196–A203 (2004).